

RANTASALMI RUKKASLUOTO, Finland

WMO: 027720

Lat: 62.063N

Lon: 28.566E

Elev: 262

StdP: 14.56

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.3	-8.4	-20.6	1.8	-15.0	-12.9	2.8	-8.2	27.2	20.9	24.8	23.4	2.3	280	0.690

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.3	78.4	67.3	75.6	66.0	72.6	64.2	69.4	76.4	67.5	73.4	65.5	71.6	8.1	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.9	99.9	73.3	64.9	93.1	72.0	62.6	85.8	69.6	33.6	76.7	32.0	73.4	30.4	71.5	75.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.1	22.4	20.2	DB	-18.5	81.8	8.4	4.3	-24.6	84.9	-29.5	87.4	-34.2	89.8	-40.4	93.0	(4)
(5)			WB	-17.7	70.7	8.5	3.2	-23.9	73.0	-28.9	74.8	-33.7	76.6	-39.9	78.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	40.6	15.8	19.5	25.4	36.1	49.3	58.5	65.0	62.2	53.5	41.8	33.9	25.4	(6)	
	DBStd	18.34	13.43	12.12	8.97	5.92	7.68	6.49	6.10	5.45	5.39	5.69	6.57	10.63	(7)	
	HDD50	4742	1060	855	761	416	107	9	0	0	29	259	484	762	(8)	
	HDD65	9017	1525	1275	1226	866	488	212	77	121	345	721	934	1227	(9)	
	CDD50	1330	0	0	0	0	86	264	464	378	134	4	0	0	(10)	
	CDD65	128	0	0	0	0	2	17	76	33	0	0	0	0	(11)	
	CDH74	425	0	0	0	0	6	48	283	88	0	0	0	0	(12)	
	CDH80	41	0	0	0	0	0	4	31	6	0	0	0	0	(13)	
	WSAvg	10.5	9.5	9.8	10.4	8.9	9.3	10.2	9.4	10.1	11.4	12.5	12.5	12.1	(14)	
	PrecAvg	26.3	2.0	1.7	1.4	1.3	1.8	2.8	3.3	3.0	2.3	2.4	2.4	2.2	(15)	
(16) Precipitation	PrecMax	32.6	3.5	3.5	3.2	2.7	5.2	5.2	5.5	4.9	5.1	4.5	5.0	3.5	(16)	
	PrecMin	20.2	0.6	0.2	0.3	0.5	0.6	0.9	1.3	0.9	0.9	0.7	0.5	0.8	(17)	
	PrecStd	2.9	0.7	0.8	0.6	0.6	1.0	1.0	1.1	1.1	1.1	1.0	0.9	0.8	(18)	
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.6	40.2	44.3	56.3	74.6	79.2	83.5	80.3	67.6	55.7	49.2	43.6	(19)	
		MCWB	36.1	36.7	37.8	46.1	59.7	64.4	69.0	69.7	60.1	52.2	46.6	42.0	(20)	
	2%	DB	35.7	36.9	40.8	51.3	69.3	74.4	79.8	76.6	64.3	53.0	45.9	39.7	(21)	
		MCWB	34.7	34.9	35.8	43.3	57.5	63.7	68.5	66.5	59.5	50.5	44.2	38.2	(22)	
	5%	DB	33.9	35.1	38.1	47.6	64.8	70.9	77.2	73.0	62.5	51.4	43.9	37.6	(23)	
		MCWB	33.0	34.1	34.5	40.7	55.5	61.9	67.2	63.9	58.4	49.2	42.7	36.2	(24)	
	10%	DB	32.5	33.6	36.1	44.4	61.0	67.9	74.3	70.3	60.6	49.5	41.6	35.9	(25)	
		MCWB	31.7	32.9	33.6	38.8	52.0	60.2	66.2	62.8	57.1	47.3	40.6	34.8	(26)	
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.7	36.9	39.0	47.3	61.7	69.4	72.0	71.5	62.0	53.3	47.1	42.1	(27)	
		MCDB	37.2	38.8	42.4	52.8	69.5	75.3	78.2	79.0	64.6	54.6	48.9	43.0	(28)	
	2%	WB	34.8	35.5	36.7	44.0	59.2	65.1	70.3	67.4	60.6	51.6	44.3	38.5	(29)	
		MCDB	35.4	36.7	39.7	49.8	67.3	71.3	77.4	73.4	63.2	52.6	45.4	39.3	(30)	
	5%	WB	33.3	34.2	35.3	41.8	56.7	62.7	68.7	65.7	59.2	49.6	42.7	36.4	(31)	
		MCDB	33.9	35.0	37.3	46.5	63.6	69.4	75.2	71.8	61.9	50.9	43.9	37.3	(32)	
	10%	WB	31.9	32.9	34.0	39.6	53.3	60.8	67.2	63.7	57.5	47.4	40.7	34.8	(33)	
		MCDB	32.6	33.7	35.6	43.3	60.0	67.3	73.2	69.1	60.0	49.1	41.5	35.6	(34)	
(36) Mean Daily Temperature Range	5% DB	MDBR	10.2	10.6	12.7	11.9	12.0	9.9	9.3	8.2	6.6	5.6	5.0	6.3	(35)	
		MCDBR	5.5	7.9	12.8	16.8	17.0	13.7	11.8	10.9	8.7	7.4	6.3	5.8	(36)	
		MCWBR	5.5	7.1	9.5	9.7	8.4	7.1	5.7	5.8	5.7	6.0	6.0	5.5	(37)	
	5% WB	MCDBR	5.2	7.8	10.0	14.9	15.5	12.7	11.2	10.0	7.9	7.0	6.5	5.6	(38)	
		MCWBR	5.4	7.2	8.2	9.3	9.1	7.0	6.3	5.9	5.8	6.4	6.2	5.5	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.242	0.291	0.318	0.358	0.354	0.350	0.375	0.359	0.336	0.317	0.261	N/A	(40)	
	taud		2.032	2.116	2.191	2.336	2.445	2.487	2.434	2.461	2.515	2.401	2.122	N/A	(41)	
	Ebn at Noon		135	207	248	263	275	278	266	261	247	203	127	N/A	(42)	
	Edh at Noon		12	22	30	32	31	30	31	28	22	17	10	N/A	(43)	
(44) All-Sky Solar Radiation	RadAvg		69	247	722	1159	1556	1684	1618	1240	701	275	72	33	(44)	
	RadStd		10	61	101	135	181	98	153	148	80	41	12	4	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (35 neighbors)	+0.92	N/A	N/A	N/A	N/A	N/A	-315	N/A	N/A	N/A

Nomenclature: See separate page